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MAJOR PROJECT REGISTRATION FORM

Date:

Project Category:	☐ Research oriented ☐ Application oriented ☐ Product oriented ☐ Other (please specify) Specify if others:	Batch ID:	AIM 07

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Title of the Project: Federated Learning for Privacy-Preserving Healthcare Diagnostics

Abstract

Building accurate AI models for healthcare diagnostics requires large volumes of diverse patient data, including medical images, clinical text, and structured records. However, individual hospitals may not have sufficient data to develop robust and generalizable diagnostic models, while sharing patient data across institutions raises significant privacy and security concerns. A common approach to address this limitation is centralized machine learning, where data from multiple hospitals is collected and trained on a central server. Although this approach can improve model performance, transferring sensitive patient data across institutions creates substantial privacy risks and limits its practical adoption in healthcare. This creates a gap between the need for large, diverse datasets and the requirement to preserve patient data privacy. Federated Learning (FL) directly addresses this gap by enabling multiple institutions to collaboratively train a shared machine learning model without transferring their raw data to a central location.Each institution trains the model locally and shares only model updates for global aggregation. FL preserves privacy by keeping sensitive patient records, including images, text, and structured data, within the institution. Thus, institutions can collaboratively develop models without sharing raw patient data. Building on this principle, the proposed solution is a Federated Learning framework that enables multiple healthcare institutions to collaboratively train a shared diagnostic model using their locally available clinical data without exchanging raw patient information. The framework primarily employs the Federated Proximal (FedProx) algorithm as its core optimization approach, where each institution performs local model training while a proximal term helps constrain local model updates toward the global model, thereby improving stability under heterogeneous and non-uniform data distributions across institutions. The locally trained model updates are then securely aggregated at a central server. The resulting federated model is evaluated against centralized and single-institution learning baselines to assess its diagnostic performance, robustness, and generalizability while maintaining patient data privacy.

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References (Category specific)

[1] McMahan, B., Moore, E., Ramage, D., Hampson, S., & Arcas, B. A. y. (2017). *Communication-Efficient Learning of Deep Networks from Decentralized Data*. Proceedings of the 20th International Conference on Artificial Intelligence and Statistics, 54, 1273–1282. PMLR paper

[2] Li, T., Sahu, A. K., Zaheer, M., Sanjabi, M., Talwalkar, A., & Smith, V. (2020). *Federated Optimization in Heterogeneous Networks*. Proceedings of Machine Learning and Systems, 2. MLSys paper

[3] Rieke, N., Hancox, J., Li, W., Milletari, F., Roth, H. R., Albarqouni, S., et al. (2020). *The Future of Digital Health with Federated Learning*. **npj Digital Medicine, 3**, 119. Nature/PubMed paper

[4] Xu, J., Glicksberg, B. S., Su, C., Walker, P., Bian, J., & Wang, F. (2021). *Federated Learning for Healthcare Informatics*. **Journal of Healthcare Informatics Research, 5**, 1–19. PubMed record

[5] Sadilek, A., Liu, L., Nguyen, D., Kamruzzaman, M., Serghiou, S., Rader, B., et al. (2021). *Privacy-first Health Research with Federated Learning*. **npj Digital Medicine, 4**, 132. Nature paper

[6] Konečný, J., McMahan, H. B., Yu, F. X., Richtárik, P., Suresh, A. T., & Bacon, D. (2016). *Federated Learning: Strategies for Improving Communication Efficiency*. arXiv:1610.05492. arXiv paper

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